Lecture 5
Stochastic Search

Armando Solar-Lezama

MCMC and Metropolis Search

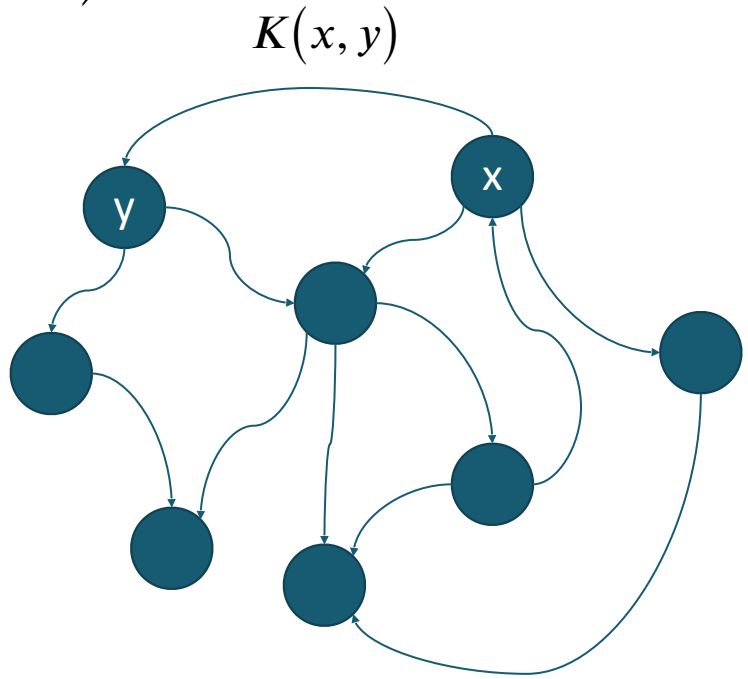
Based on “The Markov Chain Monte Carlo Revolution”
Persi Diaconis

Markov Chains

Let χ be a finite set

A Markov chain is defined by a matrix $K(x, y) : \chi \times \chi \to \mathbb{R}$

- $K(x, y) \geq 0$
- $\sum_y K(x, y) = 1$
- Probability of transitioning from x to y



Markov Chains

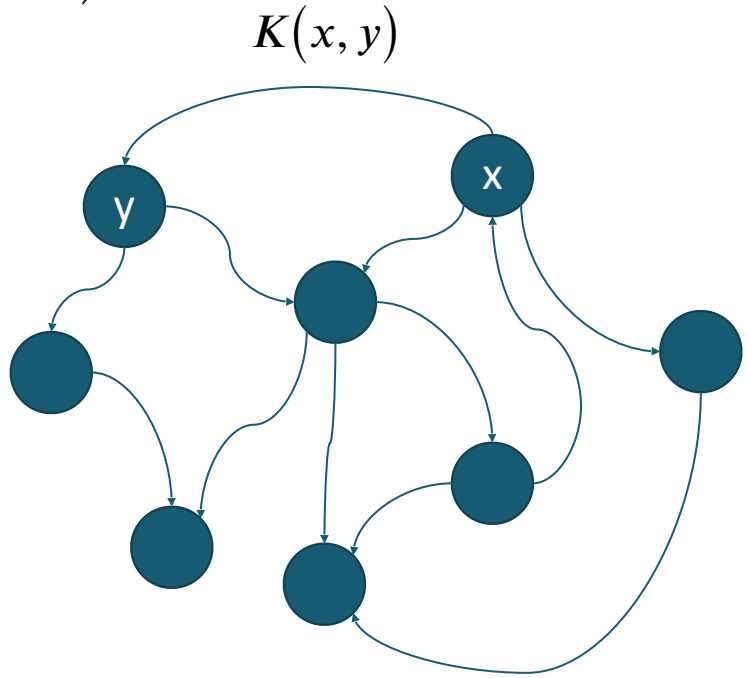
Let χ be a finite set

A Markov chain is defined by a matrix $K(x, y) : \chi \times \chi \to \mathbb{R}$

- $K(x, y) \geq 0$
- $\sum_y K(x, y) = 1$

Probability of a series $X_0,\ X_1,\ X_2 \ldots$

- $P(X_1 = y \mid X_0 = x) = K(x, y)$



Markov Chains

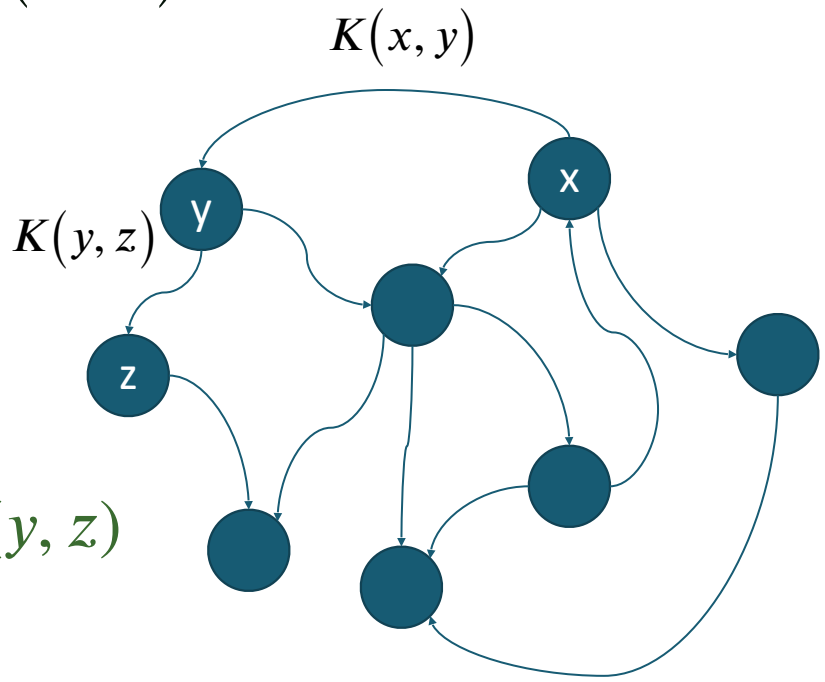
Let χ be a finite set

A Markov chain is defined by a matrix $K(x, y) : \chi \times \chi \to \mathbb{R}$

- $K(x, y) \geq 0$
- $\sum_y K(x, y) = 1$

Probability of a series $X_0,\ X_1,\ X_2 \ldots$

- $P(X_1 = y \mid X_0 = x) = K(x, y)$
- $P(X_1 = y,\ X_2 = z \mid X_0 = x) = K(x, y)K(y, z)$



Markov Chains

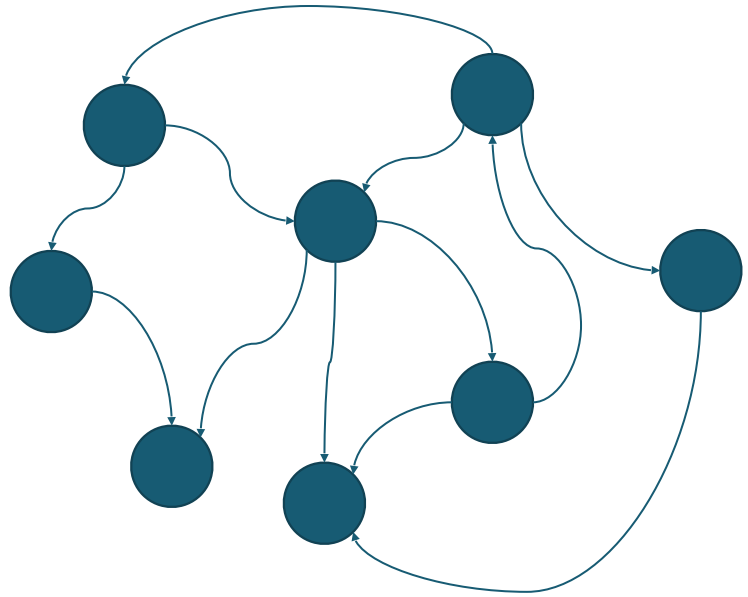
Let χ be a finite set

A Markov chain is defined by a matrix $K(x, y) : \chi \times \chi \to \mathbb{R}$

- $K(x, y) \geq 0$
- $\sum_y K(x, y) = 1$

Probability of a series $X_0,\ X_1,\ X_2 \ldots$

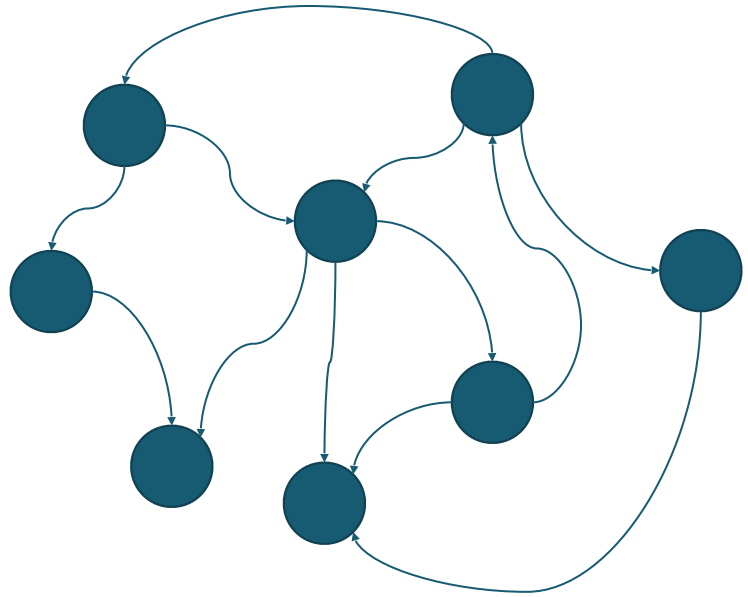
- $P(X_1 = y \mid X_0 = x) = K(x, y)$
- $P(X_1 = y,\ X_2 = z \mid X_0 = x) = K(x, y)K(y, z)$
- $P(X_2 = z \mid X_0 = x) = \sum_y K(x, y)K(y, z)$
  - This is matrix multiplication!



Stationary distribution

What is the probability $\pi(x)$ of being in a node x at some arbitrary step?

- $\pi(x) > 0 \; and \; \sum_x \pi(x) = 1$
- $\pi(y) = \sum_x \pi(x) K(x, y)$
  - i.e. $\pi = \pi K$



Fundamental theorem of (finite) Markov chains

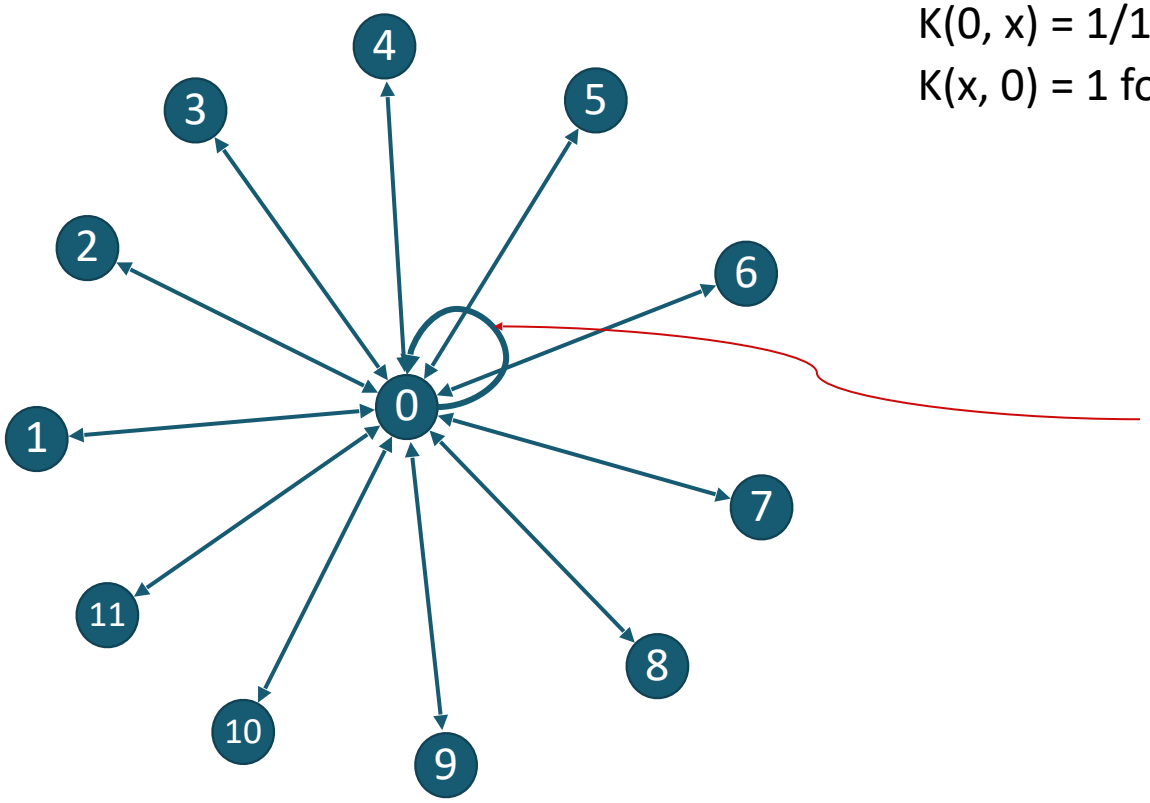
If there is an n_0 s.t. $\forall x, y . \quad n > n_0 \Rightarrow K^n(x, y) \geq 0$

- i.e. the matrix is connected.
- the matrix must also be aperiodic

$\forall x . \quad \lim_{n \to \infty} K^n(x, y) = \pi(y)$

- The n'th step of a run starting at x has probability close to $\pi(y)$ of being at y if n is large.

Stationary Distribution



K(0, x) = 1/12
K(x, 0) = 1 for x !=0

This edge makes it aperiodic
without this edge, if it starts
at the center, the probability of being
at the center on an odd step 2*n would be
zero for any n.

MCMC Based synthesis

Approach:

- Let χ be the space of programs
- Engineer a $K(x, y)$ such that $\pi(x)$ is high for “good programs” and low for “bad programs”
- Pick a random start state x_0
- Simulate the markov process for n steps for some large n.
- By the fundamental theorem, the probability of x_n is a good program will be higher than the probability that it is a bad program

Motivating example

the quick brown fox jumps over the lazy dog



yjr wiovl ntpem gpc kizqd pbrt yjr asxu fph

What is the secret function f?

Suppose we know probabilities of two letter sequences

- $\pi(f) = \frac{1}{z}\prod_i M(f(s_i), f(s_{i+1}))$

If we can sample repeatedly from π we can find the most likely function f

Metropolis algorithm

Start with a markov matrix $J(x, y)$ with $J(x, y) > 0 \leftrightarrow J(y, x) > 0$

$$K(x,y) = \begin{cases} J(x,y) & \text{if } x \neq y,\ A(x,y) \geq 1 \\ J(x,y)A(x,y) & \text{if } x \neq y,\ A(x,y) < 1 \\ J(x,y) + \sum_{z:A(x,z)<1} J(x,z)(1 - A(x,z)) & \text{if } x = y \end{cases}$$

$A(x, y)$ is the acceptance ratio $\dfrac{\pi(y)J(y,x)}{\pi(x)J(x,y)}$

Note $\pi(x)K(x,y) = \pi(y)K(y,x)$

- Then $\sum_x \pi(x)K(x,y) = \sum_x \pi(y)K(y,x) = \pi(y)\sum_x K(y,x) = \pi(y)$

Back to example

$$J(f, f^*) = \begin{cases} \frac{1}{n(n-1)(m-n+2)(m-n+1)} & \text{if } f, f^* \text{ differ in at most two places} \\ 0 & \text{otherwise} \end{cases}$$

Random switch of two symbols

- Note that $J(f, f^*) = J(f^*, f)$ so $A(f, f^*) = \pi(f^*)/\pi(f)$

Many recent synthesis applications

Influential work by Schkufza, Sharma Aiken.

- Focus on program optimization

Wide variety of applications in other areas

- Probabilistic programming
- Cognitive Science
- etc.

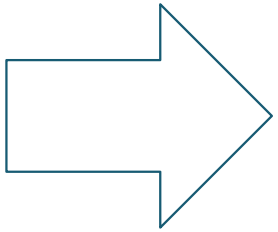
Key issues

Need good estimates of π

Where does it work?

Stochastic Superoptimization

```
 1 # gcc -O3
 2
 3 .L0:
 4   movq rsi, r9
 5   movl ecx, ecx
 6   shrq 32, rsi
 7   andl 0xffffffff, r9d
 8   movq rcx, rax
 9   movl edx, edx
10   imulq r9, rax
11   imulq rdx, r9
12   imulq rsi, rdx
13   imulq rsi, rcx
14   addq rdx, rax
15   jae .L2
16   movabsq 0x100000000, rdx
17   addq rdx, rcx
18 .L2:
19   movq rax, rsi
20   movq rax, rdx
21   shrq 32, rsi
22   salq 32, rdx
23   addq rsi, rcx
24   addq r9, rdx
25   adcq 0, rcx
26   addq r8, rdx
27   adcq 0, rcx
28   addq rdi, rdx
29   adcq 0, rcx
30   movq rcx, r8
31   movq rdx, rdi
```



```
 1 # STOKE
 2
 3 .L0:
 4   shlq 32, rcx
 5   movl edx, edx
 6   xorq rdx, rcx
 7   movq rcx, rax
 8   mulq rsi
 9   addq r8, rdi
10   adcq 0, rdx
11   addq rdi, rax
12   adcq 0, rdx
13   movq rdx, r8
14   movq rax, rdi
```

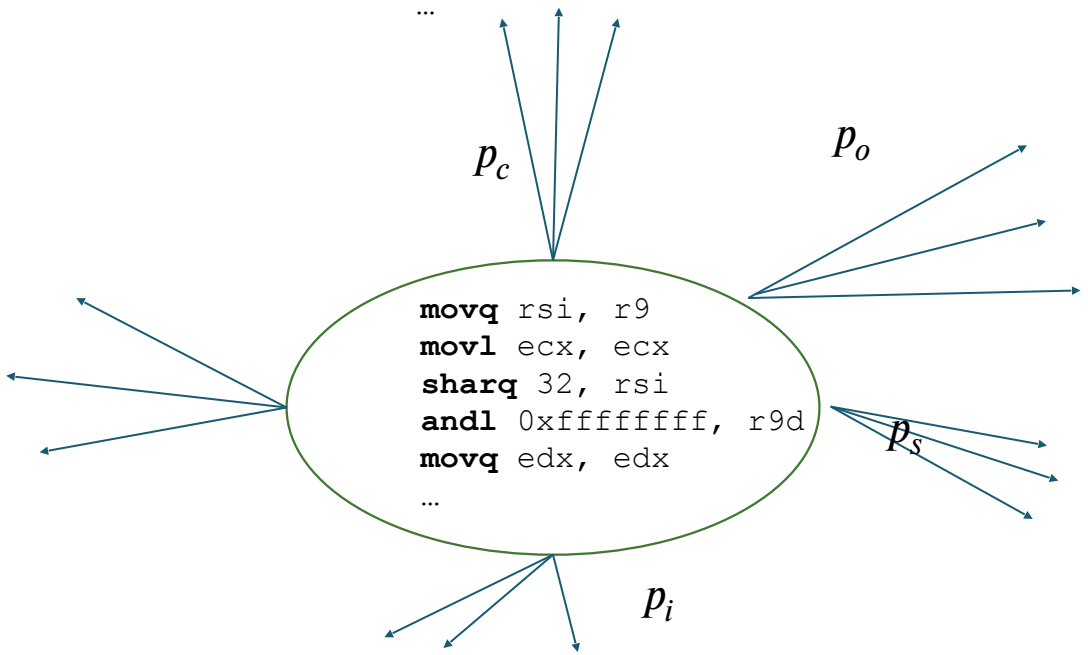
Goal: Synthesize equivalent assembly program that is significantly more efficient.

Example from Schkufza, Sharma and Aiken from ASPLOS 13.

The program space

Sequences of assembly instructions of bounded length

The proposal Distribution



Transitions that change one opcode

```
movq rsi, r9
movl ecx, ecx
sharq 32, rsi
xorl 0xffffffff, r9d
movq edx, edx
…
```

Transitions that change one operand

```
movq rsi, r9
movl ecx, edx
sharq 32, rsi
andl 0xffffffff, r9d
movq edx, edx
…
```

Transitions that replace one instruction with a NOOP

```
movq rsi, r9
movl ecx, ecx
sharq 32, rsi
NOOP
movq edx, edx
…
```

Transitions that swap two instructions

```
movq rsi, r9
movl ecx, edx
andl 0xffffffff, r9d
sharq 32, rsi
movq edx, edx
…
```

Transitions that replace one instruction with another

```
movq rsi, r9
movl ecx, ecx
sharq 32, rsi
mulq rsi
movq edx, edx
…
```

The stationary distribution

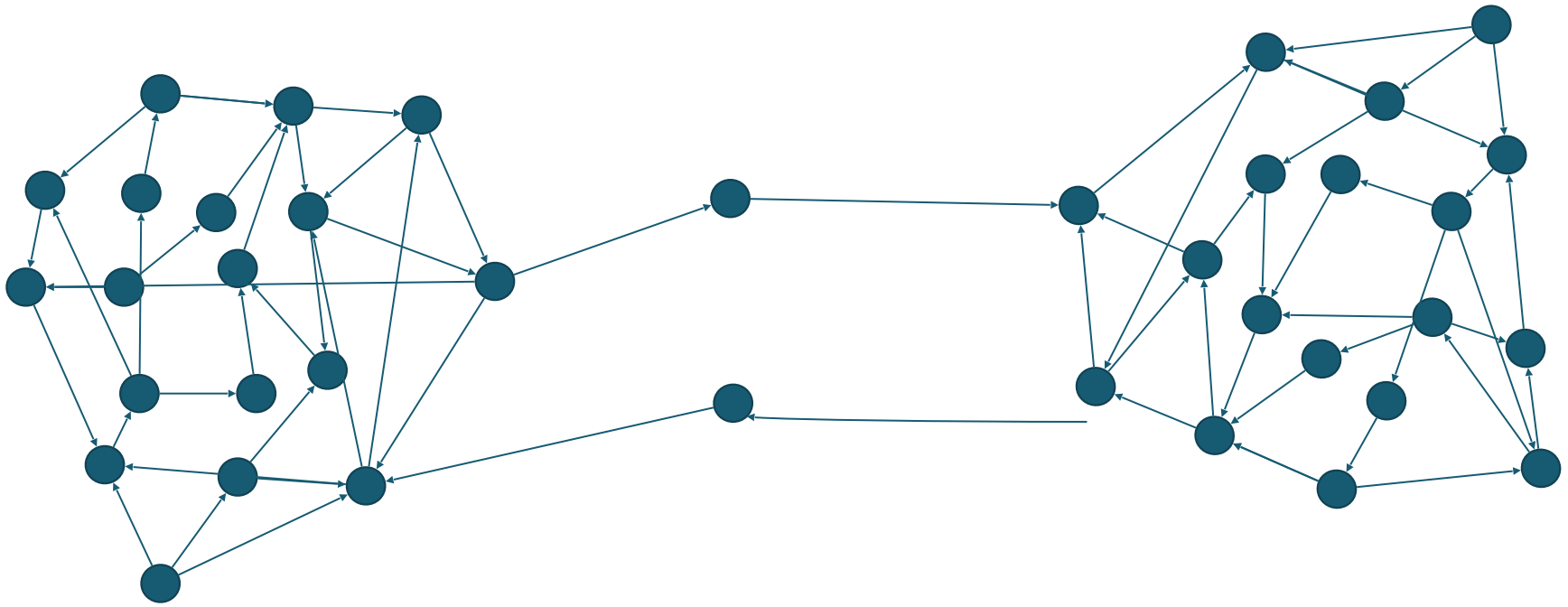
$$\pi(\mathcal{T}) = \frac{1}{Z} e^{-\beta(eq(\mathcal{R}, \mathcal{T})+perf(\mathcal{R}, \mathcal{T}))}$$

Improvements

search is conducted in two stages

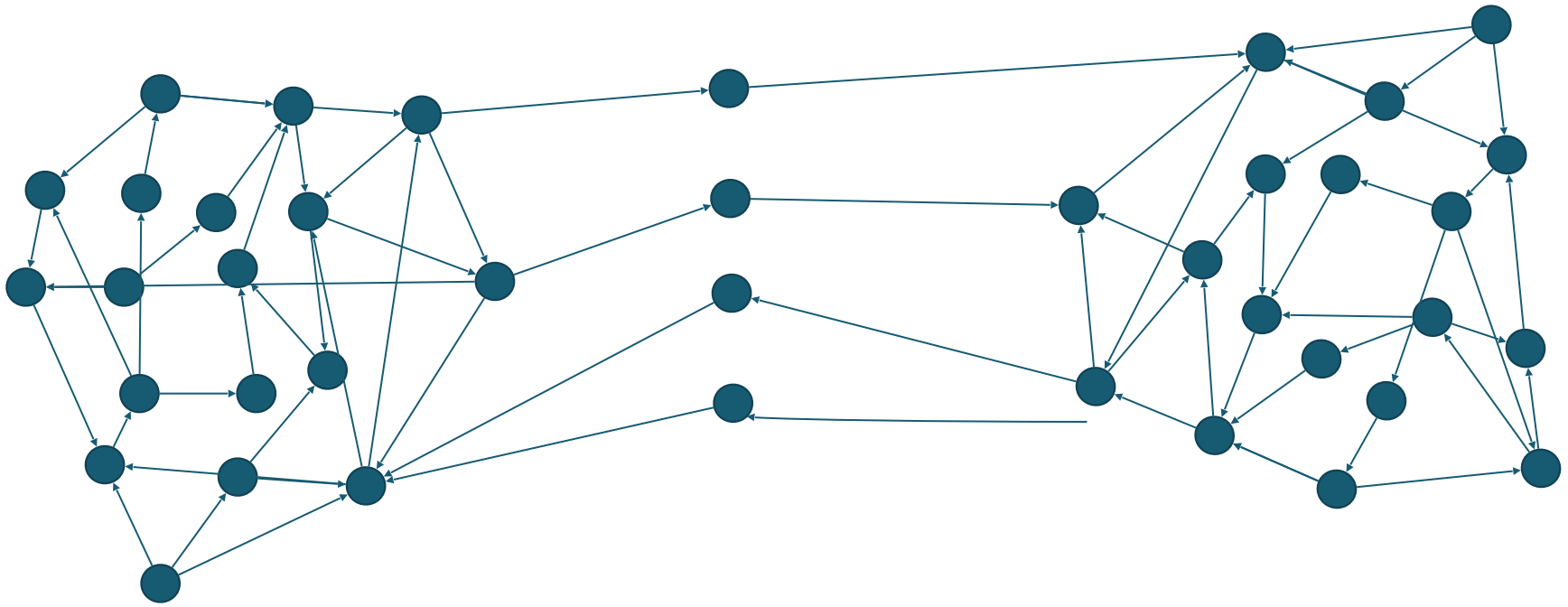
- In the first stage, the performance component is completely ignored, allowing the search to discover correct programs that are very different from the initial one.
- A set of these correct programs discovered in the first phase are used as starting points for a second phase search that includes the performance term.

Almost disjoint clusters



Starting in one cluster, the probability of transitioning to the other is extremely low

Almost disjoint clusters



Introducing more paths, or increasing the probability for the existing ones helps us converge faster to a distribution that is representative of both clusters